An SDN-based Firewall for Networks with Varying Security Requirements

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***Abstract*—With the new coronavirus crisis, medical devices' workload has increased dramatically, leaving them growingly vulnerable to security threats and in need of a comprehensive solution. In this work, we take advantage of the flexible and highly manageable nature of Software Defined Networks (SDN) to design a thoroughgoing security framework that covers a health organization's various security requirements. Our solution comes to be an advanced SDN firewall that solves the issues facing traditional firewalls. It enables the partitioning of the organization's network and the enforcement of different filtering and monitoring behaviors on each partition depending on security conditions. We pursued the network's efficient and dynamic security management with the least human intervention in designing our model which makes it generally qualified to use in networks with different security requirements.**

Keywords—SDN, security, firewalls, monitoring, Health Information Network, connected medical devices

I. INTRODUCTION

Not long after Mahler *et al.* [1] warned medical professionals about the security threats confronting connected medical devices, the world faces a pandemic that, according to several presses [2] [3], has increased the demand for and the workload of such equipment, intensifying the security risks facing them. According to [1], the cyberattacks that threaten critical medical devices, like CT scans, mostly include disruptions in the devices' functioning or results caused by manipulation, unauthorized access, and DoS attacks.

In this work, we propose an SDN-based security framework that meets a health organization's security requirements. Our proposed solution will enable systems classification based on their data and roles' sensitivity and provides access control between these systems in the network. With our model, specified parts of the network can be thoroughly isolated, and the entire network is persistently monitored and safeguarded dynamically against security threats, protecting the availability of its internal devices. [4] [5] Our security solution is indeed an advanced firewall that serves filtering and monitoring functions within the flexible framework of Software Defined Networks.

In effect, Firewalls have long been utilized as network security shields to protect networks from potential threats. However, traditional non-SDN firewalls suffer from certain drawbacks, basically caused since traditional networks have been developed to provide connectivity, and not security [6], which dissuaded us from adopting them in this work. To further elucidate, the firewall's installation and concentration on the edge of the network make it a bottleneck and a single-point-of-failure [7] that can only protect the network from external threats leaving off the internal ones. The firewall's vendor-specific and hardware/platform-dependent design limit the open extension and placement of different network products and pushes up the overall costs. [9]

In contrast to their traditional peers, Software Defined Networks provide a favorably manageable and flexible environment that enables the network-wide distribution and enforcement of security policies among network switches, creating a balance in the processing workload. Furthermore, their open design offers scalability in a less costly manner.

On the whole, we leverage SDN framework to propose a firewall solution that helps 1) distribute filtering rules throughout the network in a flexible and efficient manner, by applying relevant rules only to their corresponding switches defined based on network admin's decisions, and 2) mitigate security threats dynamically and in real-time based on the security requirements of different parts of the network. In fact, in this solution, network switches can be grouped into multiple subsets (of the network) with each subset receiving only relevant security configurations. The functionalities provided by this design makes it suitable for use in network environments which consist of parts with varying security requirements, like health organizations.

II. TECHNICAL BACKGROUND

Our proposed solution comprises a distributed SDN firewall that utilizes Access Control Lists, in the context of OpenFlow protocol, for efficient filtering and monitoring purposes to prevent or mitigate cyberattacks, such as DoS/DDoS attacks, both statically and dynamically in health organizations. The mentioned concepts are illustrated in following subsections.

A. Firewalls and Filtering

A firewall is primarily a software, hardware, or a combination of both that monitors the traffic passing through a network and drops the illegible packets. To ensure that access to the network's internal assets is exclusive to legitimate packets, the entire network traffic should, advisably, pass through the firewall. With respect to the OSI layer up to which firewalls can analyze traffic, they are classified into three categories. These include traditional packet filters, stateful packet filters, and application gateways[10].

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B. Monitoring

Monitoring refers to the periodical scanning of the network for detecting vulnerability exploits against a system. In this sense, monitoring underpins the function of Intrusion Detection Systems (IDS) and their more proactive counterparts, namely the Intrusion Prevention Systems (IPS). [11] [12]

C. Access Control Lists

Considering S as the set of subjects (which refers to active entities such as the systems in a network) and R as the set of rights, an Access Control List (ACL) l is a set of pairs presenting the features displayed in equation (1). [13]

$$l = \{(s,r): s \in S, r \subseteq R\} \quad (1)$$

D. Software Defined Networking

SDN architecture was originally derived from an idea; Casado *et al.* first proposed a network model called Secure Architecture for the Network Enterprise (SANE). A year later, the same person designed the Ethane framework. This model was proposed because of the insecure and decentralized nature of traditional networks. Later, the proposal was developed to give rise to today's SDN architecture. [14]

According to Open Networking Foundation (ONF) [15] "in the SDN architecture, the control and data planes are decoupled, network intelligence and state are logically centralized, and the underlying network infrastructure is abstracted from the applications". The layers in this architecture, from top to bottom, are as follows:

1) Application Layer: This layer is where the network's controlling applications, such as firewall and IDS, are provided in the form of network services, which, unlike traditional networks, do not need special equipment. [16]

2) Control Layer: In this layer, the actual controlling functions are realized. Here, a logically centralized controller keeps a record of network states as well as the controlling data and conveys the controlling applications' instructions to network devices. The word "logically" is used since, for scalability, availability, and geographical reasons, the controller is actually made up of distributed components that work in parallel to form a system that looks like a centralized unit in outer view. These components, in addition to different layers of the SDN architecture, communicate through secure interfaces. [16] [17] One of the most important of these communication protocols is OpenFlow which is one of the first SDN standards that enables the communication between the control and data planes.

3) Infrastructure Layer: The infrastructure layer can be divided into two sub-layers (planes): physical layer and virtual layer; the physical layer consists of the physical infrastructure of the network, and the virtual plane represents an abstraction and virtualized version of those resources. [8]

III. Related Work

Various models have been proposed for SDN-enabled filtering and security monitoring system. While some of the resources offer making filtering decisions centrally and through the controller, some others suggest enforcing security policies by adjusting each OpenFlow switch individually. Yet, others leverage some design aspects of both of the previous methods by bringing forward creative strategies to enable the localized enforcement of the network's security rules while still configuring the OpenFlow switches through the centralized controller. In addition to the papers discussing SDN firewalls, we have investigated and presented a survey of resources which propose techniques for dynamic monitoring and detection of attacks in the network.

Kao *et al*. [18] use a combination of traditional network devices and centralized SDN solutions to investigate and filter malicious traffic. In their model, packets will be delivered to an IPS Firewall through an SDN switch after passing through a network router. The IPS firewall will generate system logs based upon the analysis of the packets and convey them to an Auto-blocking server, where security decisions will be made according to the logs and advertised to the SDN controller.

Tran and Ahn [19] aim to reduce the redundancy in the network. They argue that installing all of the firewall rules on all OpenFlow switches is inefficient as an unauthorized packet will be blocked in the closest switch to where it has emanated. They also mention that blocking traffic only in one direction and permitting it to pass in the reverse order (the response to which will ultimately be blocked) brings about redundancy. Hence, they propose saving extra information about the network on the controller to selectively install each Firewall rule only on the right switches.

Bakkar *et al*. [20] propose a solution that benefits from both permissive and restrictive filtering techniques using the pipelining capability of OpenFlow switches. With their model, which has inspired our work, the network's holistic and compartmental security management is possible.

Xing *et al*. [21] combine capabilities of the popular and long-standing Intrusion Prevention System "Snort" with those of the OpenFlow framework to detect the illicit traffic in the network and reconfigure the cloud system correspondingly.

Bawany *et al* [22]. provide a technical analysis of a variety of SDN intrusion detection methods. Following that, they propose a simple and highly customizable formula, which they leverage to detect DDoS attacks in smart city applications. Their model has also inspired our work.

Our firewall solution that enables enforcing filtering rules locally and without the need to configure each switch individually. Our solution also provides dynamic monitoring and threat detection based on the real-time state of the network[1].

IV. The Proposed Firewall

In designing our proposed solution, we pursued two main goals, namely the efficient distribution of firewall rules and dynamic security management of the network in a highly controllable manner. On that account, our design consists of two major interrelated modules that serve the required functionalities. These include monitoring and distributed filtering modules. The former dynamically adjusts the security of the network, and the latter is responsible for the efficient distribution of filtering rules.

For the on-the-fly controlling and configuration of the mentioned modules, a REST Web Services Gateway Interface (WSGI) can be used, which helps adjust the network's security, flexibly, through the design's main entities.

[1] The code for our proposed firewall can be accessed at https://github.com/Ghazal-Rezaei/SDNfirewall

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In the following subsections, we will provide a description of each of the design's modules and their key objectives. Then, we will explain the main entities of the design that lead to the realization of those modules. In the final part of this section, we will put everything together to present the big picture of the design.

A. Filtering

This module is mostly responsible for the efficient distribution of firewall rules, in a way that each OpenFlow switch merely receives the filtering configurations it needs with respect to security policies of the network. Here, the word efficient indicates that we do not want every single rule to be applied on every switch in the network; Instead, we need a way of grouping OpenFlow switches to map only relevant subsets of filtering rules to the switches within each group automatically. With the functionality provided by this module, we can apply a group of rules to a batch of OpenFlow switches in the network through the logically centralized controller and without having to configure each switch directly and individually (which increases the possibility of human errors).

Through partitioning the network, the module facilitates the flexible management of the network and enables both holistic and regional distribution of filtering rules; In holistic distribution, a rule should be applied on the set of all of the network switches. In regional distribution, the rule should be applied only on the desired subset of switches. In this manner, dividing network switches into subsets is fulfilled through policy domains which will be further discussed in section IV-C. [20]

Moreover, this module's design contributes to reliable manipulation of network security-related entities; When security policies of the network change and lead to modifications in the network, like removing a filtering rule, the effects of this change should be consistently reflected throughout the entire infrastructure. For the mentioned case, the OpenFlow switches that used to apply the respective rule, should not apply it anymore. This requirement will be further illustrated in the following subsections.

B. Monitoring

This module's primary task is the periodical and persistent monitoring of the network switches' states and traffic statistics by the controller to dynamically detect and restrict anomalous traffic. Doing so, this component adapts the network security level to protect its internal hosts and OpenFlow switches from malicious traffic and DoS/DDoS attacks. A point to emphasize here is that the word dynamic indicates that this module adjusts the network security level in response to the monitored network state.

In designing this part, we used a general method of threat detection that comprehensively covers the varying requirements of different subnets while being lightweight, fast, and not introducing much overhead to the controller. For this reason, after real-time collection of network data, during specific periods, we apply a filter, presented in equation (2) [22], and inspired by the Exponential Weighted Moving Average (EWMA), to predict the expected volume of network traffic.

$$PT_t = a.PT_{t-1} + (1-a).CT_t + c,$$
$$PT_0 = CT_0 \qquad (2)$$

In the above formula, PT presents the predicted traffic; with t being the unit of time, PT_t is the expected volume of the traffic at the present time, and PT_{t-1} is that of the previous instant. CT_t is the actual volume of traffic at the current moment, derived from SDN controlled devices in the network. a is a gain or smoothing parameter which defines the sensitivity of the formula in detecting suspicious traffic by specifying the influence of the observed volume of traffic CT_t on the prediction of PT_t ; The lower this value, the faster a traffic is recognized as a threat. Finally, c is a constant determined based on the traffic characteristics. To detect the chance of traffic being threatening, the current volume of traffic is compared to the expected volume, calculated through the mentioned formula; If CT_t exceeds PT_t , it should possibly be restricted by the mitigation system.

In our proposed model, parts of the network with similar requirements, with regard to their reactivity and tolerance towards potential DoS/DDoS attacks, can be grouped into the same policy domains, having the same smoothing parameter applied to their monitored traffic. To shed more light, those regions that need to immediately respond to potential threats may be considered as highly reactive subnets (with their a closer to zero), those which can tolerate more attack risks in exchange for more accurate threat detection, with fewer false positives, and a better performance can be called intermediate reactive domains (having their a closer to one half), and those that are least sensible to DoS/DDoS attacks can be classified into the least reactive group (with their a closer to one). Much the same as the previous module, the anomaly detection sensitivity of a domain in the network can change in response to the changes in network security policies, and the impacts have to be reflected consistently.

C. Main Entities

The chief agents in this design, whose roles are elaborated upon in the following sections, are the ACL rules, policy domains, and switches, the relationships among which are as follows:

- Each rule can only be assigned to one policy domain.
- Each policy domain can have several rules and be assigned to multiple switches. [20]
- Every switch can be a member of several policy domains.

To guarantee that the security requirements are met as intended across the network and ensure reliability and consistency, whenever a modification is introduced to one of the entities, its effects must be propagated through other entities in relationship with the corresponding entity. The mentioned relationships are depicted in the Entity-Relationship Diagram in figure (1).

1) ACL Rules: They are the constraining firewall rules that aim to block malicious traffic flows. Each rule describes a specific traffic flow through five-tuple (as presented by primary keys in figure (1)) and is ultimately mapped to an entry within the flow table of an SDN switch. Also, each rule specifies the action that shall be taken when traffic is matched with the respective flow table entry. On top of that, each rule belongs only to one policy domain, which links it to a subset of network switches.

Two approaches can be taken to distribute ACL rules throughout the network;[20] Flow Table Entries (FTE),

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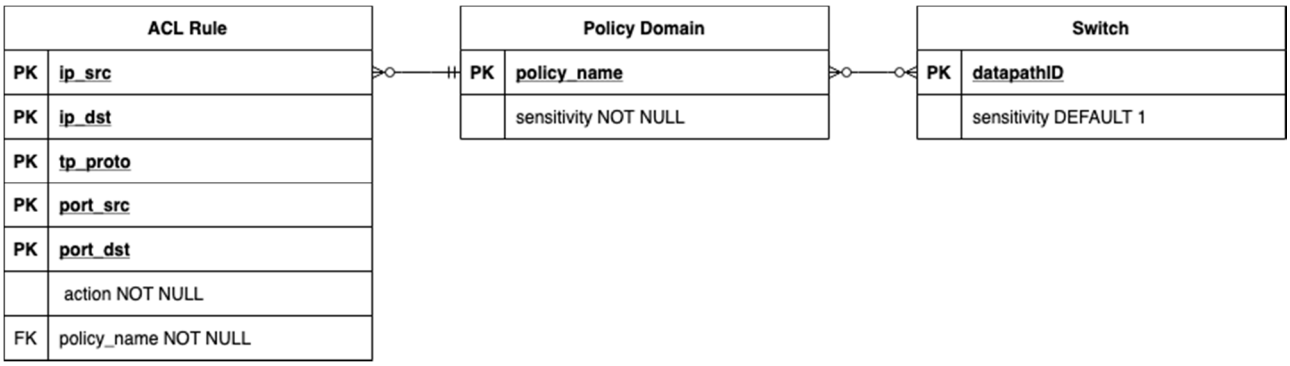


Fig. 1. ERD for the advanced firewall.

derived from the rules, can be deployed both proactively or reactively. In the proactive approach, the controller advertises the new flow table entries to the related switches as soon as they are generated. From then on, switches handle receiving flows based on their flow tables' contents and without involving the controller. In a reactive firewall, however, filtering rules are stored on the controller, and switches refer to it when getting a new packet that does not match any of the previously installed entries on their flow tables. The second approach will clearly introduce some communication and analysis overhead to the controller, making it a bottleneck in the distribution of ACL rules. Thus, the proactive approach is preferred over the reactive one in our design.

2) Policy Domain: It is an abstract concept that plays a key role in the network's flexible management by interconnecting ACL rules to OpenFlow switches; Each policy domain contains a group of rules, which will be assigned only to the network switches belonging to that particular policy domain (The relationship between policy domains and network switches is further elaborated in figure (2)). In this respect, a policy domain can be likened to an ACL (as described in section II-C), with the set of filtering rules(R) similar to the rights in the ACL definition and the set of network OpenFlow switches(S) resembling the subjects in the same definition. According to the mentioned definition, the effect of changes can be readily propagated among related entities. Below are the possible cases:

- Suppose a new ACL rule is created and attributed to a particular policy domain. In this case, all of the OpenFlow switches belonging to that domain will apply that rule within their respective subnets. Likewise, if an ACL rule is revoked from a policy domain, its related FTE will also be removed from the flow tables of the member switches.
- When a switch is registered within a policy domain, all of the ACL rules belonging to that policy domain will be assigned to it. Similarly, if a switch is excluded from a policy domain, all related rules will be revoked from it.

Additionally, there is a level of sensitivity related to each policy domain, which defines the desired gain(a) to be applied to the member switches of the domain in the formula used for the monitoring module's anomaly detection function. The DoS/DDoS detection sensitivity associated with the network OpenFlow switches is chosen based upon the policy domains they belong to.

3) Switch: This entity is directly mapped to the actual OpenFlow switches of the network whose information, such as unique identifiers and instances of the OpenFlow protocol, is stored on the controller. In our design, every SDN switch in the network may have special security requirements, which entails assigning distinct filtering rules and a unique gain/sensitivity (to use in anomaly detection formula) to it. This implies a switch may belong to different policy domains, all with their unique rules, in accordance with the network security policies. The policies defining the ACL rules and the gain that must be applied to a given switch can be developed based on different metrics such as the location of the switch in the network topology, its type and technology, its traffic threshold, etc.

If a switch is a member of multiple policy domains, a union of their respective rules will be applied on its subnet, and the lowest value among the smoothing parameters (the a that introduces the greatest sensitivity) associated with each of these policy domains will be chosen to use in the anomaly detection filter. Since, depending on the changes of their policy domains, switches may be assigned varying a values, in the monitoring module, the amount of this value is not cached and will be checked every time the statistics related to a switch are retrieved to use in the introduced formula. The discussed changes can affect the value of a in one of these two ways:

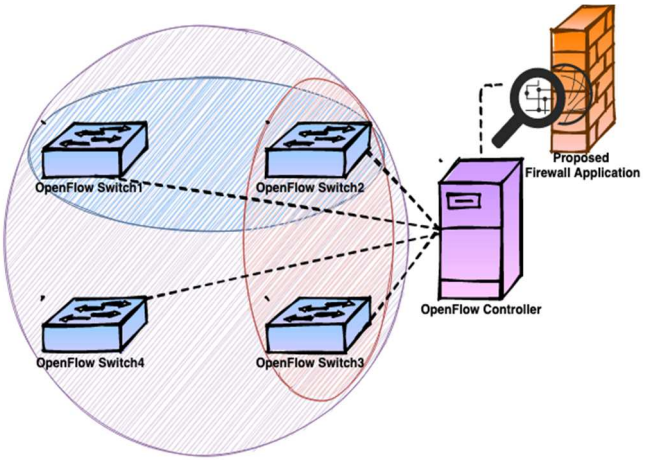


Fig. 2. Policy domains (Ellipses) attributed to switches in a network. As observable, a policy domain can be assigned to multiple switches, and each switch can be a member of more than one policy domain.

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- If a switch is registered within a new policy domain, its new a value will be the lowest among all of its previous policy domains together with the new one.
- If a switch is excluded from a policy domain, its new value of a will be the lowest among all of its remaining policy domains.

D. Putting Everything Together

With our proposed solution, different OpenFlow switches in the network can be grouped together, based on their common features and with respect to the network admin's decision, to form subsets of the network. Each subset of switches is mapped to a policy domain which provides the means to 1) the efficient distribution of filtering rules, and 2) the flexible and dynamic security management of the network; For the efficient assignment of filtering rules, each group of rules is only applied to a subset of network switches provided that both groups (the ACL rules and the switches) belong to the same policy domain. For the dynamic adjustment of the network security level, each policy domain is given a gain (a) that defines its desired responsiveness towards potential DoS/DDoS attacks. The same amount of a is applied to all of the switches within that policy domain to make sure they are at least as responsive. When a switch is a member of multiple policy domains (as is the case with three of the switches in figure (2)), it will apply a combination of all of the rules in those policy domains to its connected devices and will choose the lowest value of a among the values associated with the mentioned policy domains to ensure the highest sensitivity and responsiveness towards security threats.

In addition, in this design, we have devised means to coherently propagate the changes throughout the network; These changes include the creation/removal of ACL rules, the creation/removal of policy domains and their assignment to/revocation from OpenFlow switches the effects of which is spread through related entities.

V. Functional Evaluation in The Target Environment

In this part, we provide evidence on why our proposed design meets the intended goals and corresponds to a health organization's security requirements. Because our system is comprised of two major modules, as described in section IV, we initially present the results in the form of unit tests and then evaluate the functionality of the entire system. [24]

A. Test Environment

To test our implementation against functional requirements, we used RYU as our controller and generated our topologies in a virtual environment. [25] Mininet was used to "create a realistic virtual network, running real kernel, switch and application code, on a single machine (VM)." [26] We leveraged ICMP messages in this environment's command line interface to generate test packets and manipulated the network via the WSGI interface, discussed in section IV. In the topology, presented in figure (3) and pertaining to a health center, we have created four policy domains in general, each of which subsumes a part of the network, as described below:

1) Edge policy domain: This slice of the network is where the organization's public servers (and generally speaking DMZ), such as the online queue management system, are located on one side. Also, The OpenFlow switch(s) of this part is located at the edge of the network and, on the other side, faces the internet directly where people and external hosts can make use of the center's online services. To enable public access and ensure good performance, the filtering and monitoring policies in this domain are less strict than other domains.

2) Untrusted Internal policy domain: Internal individuals, who can potentially be malicious, connect to the OpenFlow switches in this domain. These individuals may need internet connection services, and the office automation and Health Information Exchange (HIE) tasks are also performed here. Therefore, in designing this policy domain, we devised more stringent filtering and monitoring strategies than the Edge to protect the internal users from prospective threats while ensuring the rendered services' quality.

3) Trusted Internal policy domain: This is the most critical part of the health center, where medical devices, such as imaging tools, are connected. Due to the irrecoverable ramifications of cyberattacks on this domain's internal devices, it requires the most rigorous security policies to minimize the possibility of such attacks.

4) Common policy domain: This policy domain, which covers the entire network, is designed to enforce security policies and rules common to all parts of the network.

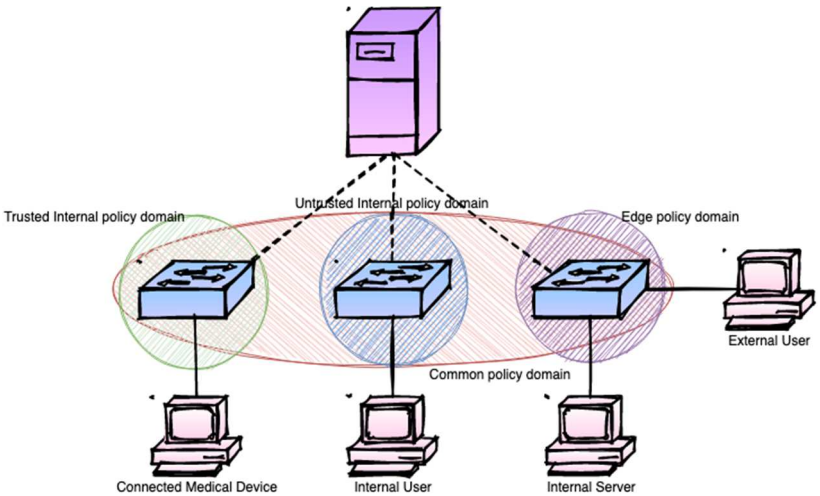


Fig. 3. The health center network topology used for testing.

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B. Evaluation of Filtering Functionality

At this juncture, we measure the filtering module's functionality against the requirements mentioned in section IV-A using the tests generated in the context of the scenario in figure (3).

In the beginning, when no ACL rules have been created and assigned to any of the policy domains, access is unrestricted throughout the entire network. In this case, all hosts can ping each other without any of the packets dropped.

In the second stage, suppose we want to restrict access from the external world to the Trusted Internal domain to protect it from external threats; For this purpose, we only need to create a new ACL rule constraining the external hosts' access within the Edge policy domain. When checking the connections between hosts, as expected, the external host cannot connect to the Trusted domain, while other hosts can still connect, and the overall packet drop rate is equal to 16%. As the mentioned security policy is automatically applied only to the desired subnet, this test confirms our solution's competence in sectional management of the network.

In the third step, assume that we need to make the Trusted Internal subnet isolated to protect it from both internal and external malicious individuals; Therefore, we create an ACL rule in the universal Common policy domain, which prohibits any access to the mentioned subnet. On this spot, predictably, the hosts in the Edge and Untrusted Internal policy domains cannot connect to the host in the Trusted Internal domain, leaving the packet drop rate to 50%. Since the required security policy is globally enforced in all domains, this test corroborates our model's capability in the network's holistic management.

To check whether the effects of the changes are congruously reflected among relevant entities, we tried the possible scenarios described in part IV-C; For instance, we tried removing the Common policy that was used in the previous part; Expectedly after then, all of the hosts save for the external one, which was denied access to the Trusted Internal domain by the ACL rule in the Edge policy domain, could connect, and the packet drop rate returned to 16%. This instance proves that the effect of deliberate security changes is propagated throughout the related entities in the network, without interference with the work of other policy domains.

TABLE I. SUMMARY FOR TESTING THE FILTERING MODULE

Network state	Packet drop rate
The initial network without any filtering rules	0%
Restricting the external hosts' access to The Trusted Internal domain	16%
Isolating the Trusted Internal domain	50%
Removing the Common domain	16%

All things considered about the filtering module, a summary of which is observable in table I, it should also be noted that we did not evaluate the matching behavior/speed of the firewall rules in the OpenFlow switches. Since we have merely written a controlling application (which distributes the filtering rules to OpenFlow switches and puts the burden of deciding about network flows on them, leaving the controller free from decision-making), installed it on the controller, and don't have any mastery on how and at what rate the data plane infrastructure will function. Thus, after ACL rules are created and shared in the form of FTEs, the end-to-end delay of traffic depends entirely on the forwarding rate of OpenFlow switches.

C. Evaluation of Monitoring Functionality

This part is dedicated to the verification of the monitoring module against the predefined objectives, explained in section IV-B. To carry out this task, each of the test network's policy domains are assigned a specific value of a which is compatible with their subnets' required traffic threshold; In the first place, because the connected medical devices in the Trusted Internal policy domain are most sensible to Denial of Service attacks, they are categorized into the highly reactive group, discussed in section IV-B, and assigned the value of 0.1 for their a. The next policy domain, namely the Untrusted Internal, is classified as intermediate reactive. Since the internal non-medical devices and individuals within this domain are not much tolerant of DoS/DDoS attacks, and, at the same time, the quality of the service rendered to them cannot be sacrificed for their security the $a = 0.5$ is a suitable value for this region. The Edge policy domain cannot stand unnecessary restrictions on the traffic that passes through it as they will adversely affect the performance of the health center's internal servers. Consequently, to minimize the chance of false negatives, we classify this subnet in the least reactive category, assigning $a = 0.9$ to it. The last one i.e. the Common policy domain, which encompasses the entire network switches, is categorized in the least reactive group having its $a = 0.8$. With the suggested configurations, it is expected that every switch in each policy domain takes the lowest value of a among the values linked to its policy domains; Thereby, the switch(s) in the Trusted Internal policy domain will take the value of 0.1 among the values 0.1 (associated with the Trusted Internal policy domain) and 0.8 (assigned to the global Common policy domain). Similarly, the switch(s) in the Untrusted Internal policy domain will choose the value of 0.5 between the values 0.5 (associated with the Untrusted Internal policy domain) and 0.8. Finally, the switch(s) in the Edge policy domain will take the value of 0.8 among the values 0.9 (associated with the Edge policy domain) and 0.8.

To test the effectiveness of the monitoring module, we flooded a host in each of the policy domains with ICMP packets; Predictably, the lower the value of a associated with a policy domain, the faster the mitigation mechanism was triggered in response to the flood, without intervention with other domains. This observation evidently proves the flexibility of the monitoring module in adjusting the network dynamically and compartmentally.

To check whether the effect of changes is accurately reflected, similar to the evaluation of the filtering module, we removed the Common policy domain. Following this, the value of a associated with the Edge policy domain was reset to 0.9, and the other policy domains remained intact. The network states before and after this change are depicted in table II).

To check how the monitoring module affects performance, we compared the network's behavior with and without this module, in terms of min/max/average RTT; The results displayed negligible difference, confirming that the module does not introduce much overhead to the network. Nevertheless, if the network data collection frequency

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increases, despite higher precision, the total overhead may grow, respectively. Thus, the tradeoff between performance and accuracy should be considered when deciding about the length of network sampling periods in the monitoring module.

TABLE II. SUMMARY FOR TESTING THE MONITORING MODULE

Network state	a value of Switch 1	a value of Switch 2	a value of Switch 3
4 available policies	0.8	0.5	0.1
3 available policies	0.9	0.5	0.1

D. Integrated Evaluation

To analyze the entire system's performance, we had both of the previous modules function in parallel; Thanks to policy domains, the firewall rules could still be distributed both globally and regionally. Simultaneously, the same policy domains facilitated dynamic attack detection and mitigation in each region. Moreover, the modifications to the main entities in the network affected every module fittingly. On the whole, the system functioned consistently, providing a high level of flexibility and manageability.

VI. CONCLUSION AND FUTURE WORK

The new pandemic has intensified the need for medical imaging devices such as CT scans, making their security threats seem more serious than ever. This incident has led us to design a firewall solution that provides security in health information networks. Our proposed solution uses the capabilities of SDN to pursue goals such as flexibility, reliability, dynamism, and automatic manageability in network security enforcement. It allows a configuration of network switches in which different subsets of switches can exhibit varying filtering and monitoring behaviors while still providing the means to the holistic configuration of the network. The strength of our model is supported by the simple tests presented in the evaluation chapter.

Within the foreseeable future, we aim to extend our design to the application layer of the OSI model by creating authentication and authorization systems that, more effectively, improve security in the organization. Moreover, in SDN networks, various controlling applicationscontrol the network simultaneously. To be more specific, network settings related to different applications may interfere in the control plane. For example, according to Hua *et al.* [26], some applications, such as load balancing programs, may inadvertently make changes in the network, such as changing some flow features or creating dependencies between rules, which can bypass firewall rules. Therefore, in later versions, we try to make the proposed firewall work properly parallel to other applications. To do so, other capabilities need to be considered, such as tracking flows in the network.

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